THE RESTRICTED MOR PRESHEAF OF A FIBRED CATEGORY IS A PRESHEAF

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Pseudo functor property (d). categories-definition-functor-into-2-category (cross-chapter, named). Justifies the second equality: the two pullback constraint maps compose by property (d) since f' = f o g.

Source reference. Original reference 003N: focused-seeds/geometry/S9-stk-painful/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *This actually does give a presheaf.*

Proof. Let $g : V'/U \to V/U$ be as above and similarly $g' : V''/U \to V'/U$ be morphisms in $\mathcal{C}/U$. So $f' = f \circ g$ and $f'' = f' \circ g' = f \circ g \circ g'$. Let $\phi \in \mathrm{Mor}_{\mathcal{S}_V}(f^*x, f^*y)$. Then we have

$$\begin{aligned} & (\alpha_{g\circ g',f})_y^{-1} \circ (g \circ g')^*\phi \circ (\alpha_{g\circ g',f})_x \\ = \; & (\alpha_{g\circ g',f})_y^{-1} \circ (\alpha_{g',g})_{f^*y}^{-1} \circ (g')^*g^*\phi \circ (\alpha_{g',g})_{f^*x} \circ (\alpha_{g\circ g',f})_x \\ = \; & (\alpha_{g',f'})_y^{-1} \circ (g')^*(\alpha_{g,f})_y^{-1} \circ (g')^*g^*\phi \circ (g')^*(\alpha_{g,f})_x \circ (\alpha_{g',f'})_x \\ = \; & (\alpha_{g',f'})_y^{-1} \circ (g')^*\Big((\alpha_{g,f})_y^{-1} \circ g^*\phi \circ (\alpha_{g,f})_x\Big) \circ (\alpha_{g',f'})_x \end{aligned}$$

which is what we want, namely $\phi|_{V''} = (\phi|_{V'})|_{V''}$. The first equality holds because $\alpha_{g',g}$ is a transformation of functors, and hence

$$\begin{array}{ccc} (g \circ g')^*f^*x & \xrightarrow{(g\circ g')^*\phi} & (g \circ g')^*f^*y \\ {\scriptstyle (\alpha_{g',g})_{f^*x}}\downarrow & & \downarrow{\scriptstyle (\alpha_{g',g})_{f^*y}} \\ (g')^*g^*f^*x & \xrightarrow{(g')^*g^*\phi} & (g')^*g^*f^*y \end{array}$$

commutes. The second equality holds because of property (d) of a pseudo functor since $f' = f \circ g$ (see Categories, Definition 003N). The last equality follows from the fact that $(g')^*$ is a functor. □

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